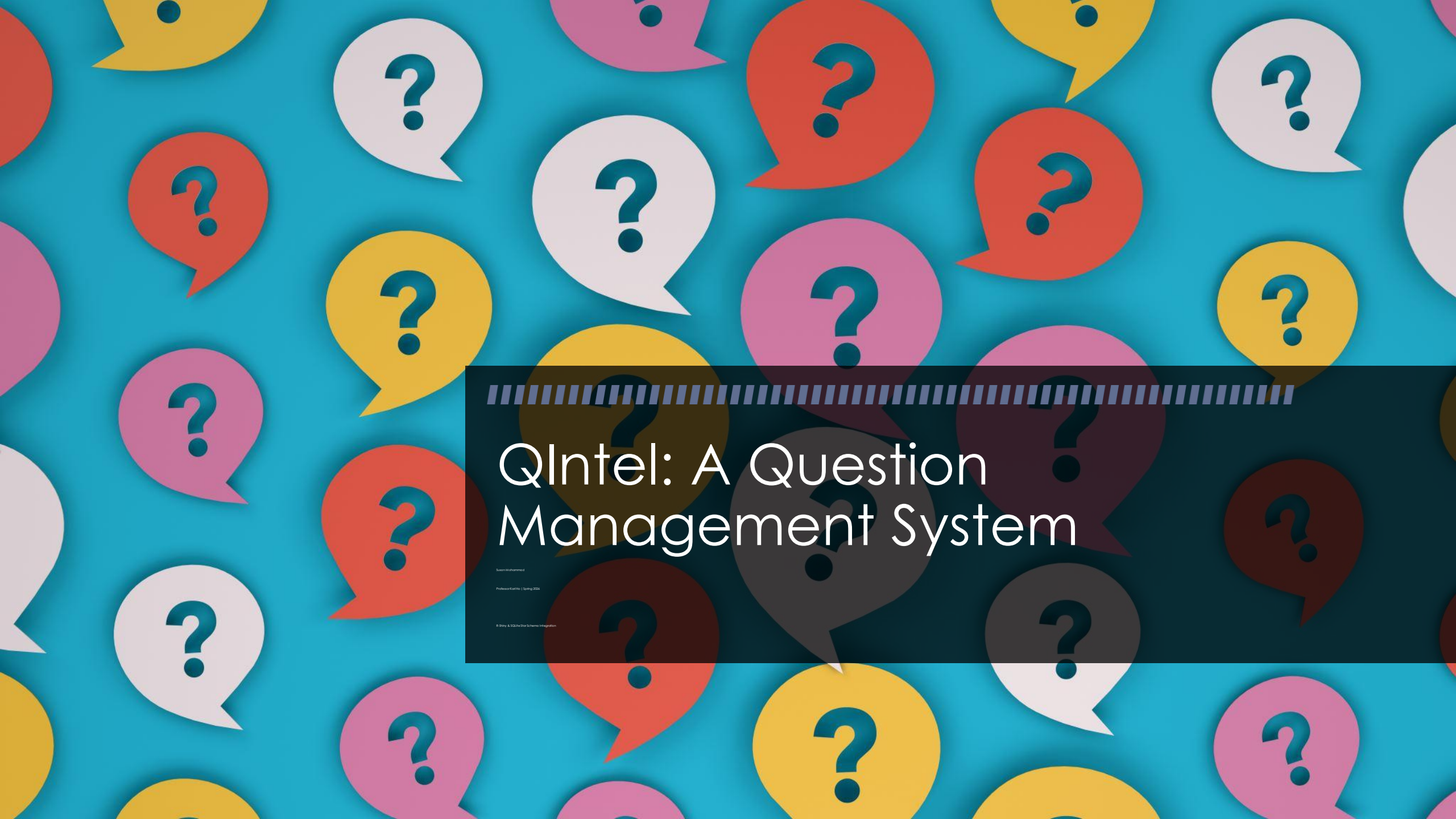




QIntel: A Question Management System

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Professor Curfio | Spring 2016

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Democratizing Access in High-Stakes Events



Press briefings and panel events often face structural chaos. Limited time frequently allows only a subset of reporters to ask questions, leading to dissatisfaction and potential bullying of speakers.



QIntel replaces chaotic situation with an organized, data-driven platform for fair question management.

Three Core Roles

The Admin

Defines core event parameters: Name, Date, and Location.

The Reporter

Submits real-time questions, identifying their organization. Data is automatically cleaned and classified by AI.

The Speaker

Review questions via the Dashboard, manages the approval queue, and engages with voice-to-text response tools.

Technology Stack for Real-Time Logic



Frontend/Backend: R Shiny for responsive event logic.



Database: SQLite with a Star Schema backend for relational integrity.



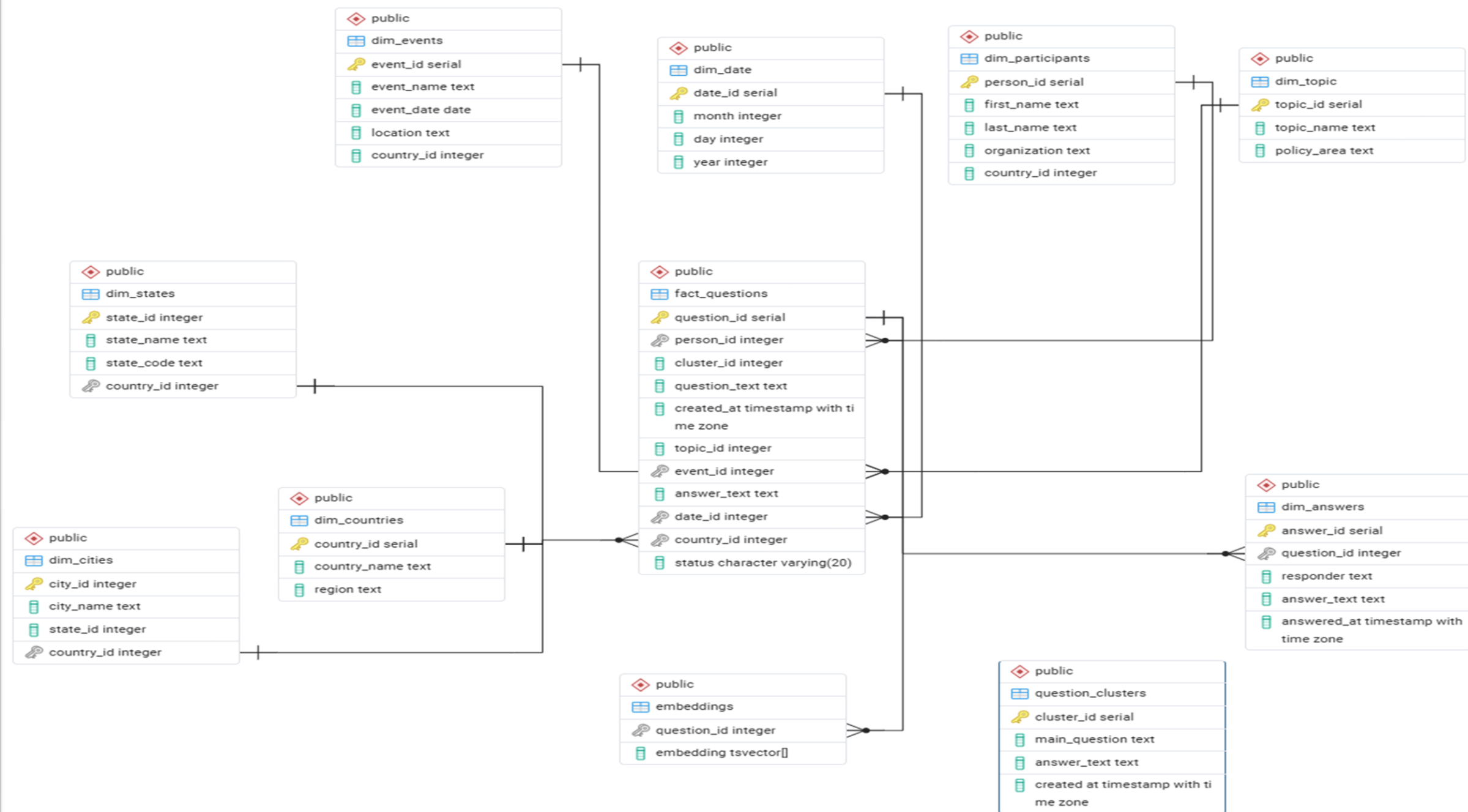
Topic Modeling: Quanteda + LDA for automated question classification.



Entity Detection: Fuzzy string matching (adist) for location/country detection.



Voice Interface: Web Speech API for seamless speaker-to-display transcription.



Database Strategy

Database Structure		
Browse Data		
Edit Pragmas		
Execute SQL		
Create Table Create Index Modify Table Delete Table Print Refresh		
Name	Type	Schema
Tables (11)		
dim_answers		CREATE TABLE dim_answers (answer_id INTEGER PRIMARY KEY, question_id INTEGER, responder TEXT, answer_text TEXT, answered_at TIMESTAM
dim_cities		CREATE TABLE dim_cities (city_id INTEGER PRIMARY KEY, city_name TEXT, state_id INTEGER, country_id INTEGER, FOREIGN KEY(state_id) REFER
dim_countries		CREATE TABLE dim_countries (country_id INTEGER PRIMARY KEY, country_name TEXT UNIQUE, region TEXT)
dim_date		CREATE TABLE dim_date (date_id INTEGER PRIMARY KEY, day INTEGER, month INTEGER, year INTEGER)
dim_events		CREATE TABLE dim_events (event_id INTEGER PRIMARY KEY, event_name TEXT, location TEXT, country_id INTEGER, date_id INTEGER, FOREIGN I
dim_participants		CREATE TABLE dim_participants (person_id INTEGER PRIMARY KEY, first_name TEXT, last_name TEXT, organization TEXT, country_id INTEGER, FO
dim_states		CREATE TABLE dim_states (state_id INTEGER PRIMARY KEY, state_name TEXT, state_code TEXT, country_id INTEGER, FOREIGN KEY(country_id) R
dim_topic		CREATE TABLE dim_topic (topic_id INTEGER PRIMARY KEY, topic_name TEXT, policy_area TEXT)
embeddings		CREATE TABLE embeddings (question_id INTEGER PRIMARY KEY, embedding TEXT, FOREIGN KEY(question_id) REFERENCES fact_questions(questio
fact_questions		CREATE TABLE fact_questions (question_id INTEGER PRIMARY KEY, person_id INTEGER, cluster_id INTEGER, question_text TEXT, created_at TIME
question_clusters		CREATE TABLE question_clusters (cluster_id INTEGER PRIMARY KEY, main_question TEXT, answer_text TEXT, created_at TIMESTAMP)
Indices (0)		
Views (0)		
Triggers (0)		



Automated Classification via LDA ModelingTopic Identification Logic

- To provide a free, locally-hosted alternative to Cloud LLMs, QIntel implements Latent Dirichlet Allocation (LDA) through the Quanteda framework.
- Fallback logic assigns "General" to outliers.

Fuzzy Location Detection & Entity Logic

- **The Levenshtein Logic**
- Detecting entities in natural language queries is prone to human error. QIntel uses adist fuzzy matching with a distance threshold ($d \leq 2$) to correct typos in real-time.
- *"United States" recognized from "USA", "US", or individual US States.*
- **Automated Seeding:** Maps package ISO3166 data populates dim_countries.
- **State Logic:** 50 US States pre-loaded for regional identification.
- **Fuzzy fallback:** Returns "Unknown" if detection fails, ensuring system uptime.



Operational Workflow & Queue Management

Submit

Reporter enters
text, AI runs
detection.

Queue

Speaker selects
multiple
questions for
the live list.

Answer

Real-time
Speech-to-Text
records
responses.

Archive

Finalized
answers saved
to dim_answers
database.

Real-Time Interaction Features



- The **Live Display** page is designed for focused interaction. Speakers can focus on one question at a time, eliminating the cognitive load of a cluttered interface.
- **Speech-to-Text Utility:**
- Leverages Web Speech API for hands-free transcription of live answers directly into the record.

Overcoming Technical Roadblocks

- **Spelling Challenges**

- Identified that Hunspell suggestions work best in a browser environment. Optimized local view to prevent detection lag.

- **AI Constraints**

- Transitioned from Cloud LLM to free Quanteda/LDA to maintain data sovereignty and zero-cost operation.

- **Flow Logic**

- Developed custom queue management to prevent "Information Overload" during the live display phase.

Thank You
Questions &
Discussion

QIntel: Susan Mohammad
Spring 2026 Information
Management

Application:

https://qevent.shinyapps.io/qevent_app/